Retail Distributions

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```
# ============================================
# Program: retail_data_vis.R
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# Date: 2026 10 27
# Description: Produce data visualization 2 - retail counts for different subsets of zips
# ============================================
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------------------------------ tidyverse 2.0
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------------------------------ tidyverse_conflicts
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(rprojroot)
library(RColorBrewer)

setwd(find_root(is_git_root))

pop <- read_csv("0_data/1_clean/base_population.csv") |>
    filter(central) |>
    mutate(
        cbsa_code = as.character(cbsa_code),
        year = as.character(year)
    ) |>
    select(
        count_totartsentertainment,
        count_coffeeshopscafes,
        count_delis,
        count_fastfood,
        count_dollarstores,
        cbsa_name,
        cbsa_code,
        year,
        low_income,
        high_income,
        zcta,
        zipcode,
        zip_rank
```

```
    ) |>
    # TODO: move this to retail cleaning code
    pivot_longer(
        starts_with("count"),
        values_to = "count",
        names_to = "category",
        names_prefix = "count_"
    ) |>
    mutate(
        category_pretty = case_when(
            category == "coffeeshopscafes" ~ "Coffee Shops",
            category == "delis" ~ "Delis",
            category == "dollarstores" ~ "Dollar Stores",
            category == "fastfood" ~ "Fast Food",
            category == "totartsentertainment" ~ "Arts and Entertainment"
        )
    )
```

```
## Rows: 5277 Columns: 36
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (3): zipcode, zcta, cbsa_name
## dbl (28): statefips, year, cbsa_code, zip_median, zip_rank, annual_avg_zhvi,...
## lgl  (5): central, gentrifiable, low_income, high_income, gentrifying
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# TODO: control for cbsa-year population size or number of total businesses in cbsa-year
↪  to make this more accurate

grouped_lo <- pop |>
    filter(low_income) |>
    group_by(cbsa_name, year, category_pretty) |>
    summarize(count = mean(count)) |>
    ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by cbsa_name, year, and category_pretty.
## i Output is grouped by cbsa_name and year.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(cbsa_name, year, category_pretty))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
summary(grouped_lo)
```

```
##      cbsa_name             year     category_pretty      count
##  Length   :30   Length   :30   Length   :30     Min.   : 0.4848
##  N.unique : 6   N.unique : 1   N.unique : 5     1st Qu.: 2.1746
##  N.blank  : 0   N.blank  : 0   N.blank  : 0     Median : 4.2699
##  Min.nchar: 6   Min.nchar: 4   Min.nchar: 5     Mean   : 4.4487
##  Max.nchar:15   Max.nchar: 4   Max.nchar:22     3rd Qu.: 5.7500
##                                                 Max.   :13.9021
```

```r
grouped_hi <- pop |>
    filter(high_income) |>
    group_by(cbsa_name, year, category_pretty) |>
    summarize(count = mean(count)) |>
    ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by cbsa_name, year, and category_pretty.
## i Output is grouped by cbsa_name and year.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(cbsa_name, year, category_pretty))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
summary(grouped_hi)
```

```
##      cbsa_name          year     category_pretty     count
##  Length   :30   Length   :30   Length   :30     Min.   :0.000
##  N.unique : 6   N.unique : 1   N.unique : 5     1st Qu.:0.886
##  N.blank  : 0   N.blank  : 0   N.blank  : 0     Median :2.675
##  Min.nchar: 6   Min.nchar: 4   Min.nchar: 5     Mean   :2.807
##  Max.nchar:15   Max.nchar: 4   Max.nchar:22     3rd Qu.:4.357
##                                                 Max.   :8.400
```

```r
grouped_med <- pop |>
    filter(zip_rank > 50, zip_rank < 75) |>
    group_by(cbsa_name, year, category_pretty) |>
    summarize(count = mean(count)) |>
    ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by cbsa_name, year, and category_pretty.
## i Output is grouped by cbsa_name and year.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(cbsa_name, year, category_pretty))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
lo_chart <- ggplot(
    grouped_lo,
    aes(y = cbsa_name, x = count, fill = category_pretty)
) +
    geom_bar(stat = "identity", position = "fill", width = 0.5) +
    xlab("Retail Distribution") +
    ylab("CBSA") +
    labs(
        title = "Figure 2: Low-Income Retail Distribution by CBSA in 2011",
        caption = "Zip codes with a median income in the \n bottom fortieth percentile
        ↪  relative to the CBSA in 2011."
    ) +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank(),
        text = element_text(size = 11, family = "Times")
```

```
    ) +
    scale_fill_brewer(palette = "Spectral") +
    guides(fill = guide_legend(title = "Retail Category"))

hi_chart <- ggplot(
    grouped_hi,
    aes(y = cbsa_name, x = count, fill = category_pretty, width = 0.5)
) +
    geom_bar(stat = "identity", position = "fill") +
    xlab("Retail Distribution") +
    ylab("CBSA") +
    labs(
        title = "Figure 4: High-Income Retail Distribution by CBSA in 2011",
        caption = "Zip codes with a median income in the top \n twenty-fifth percentile
        ↪  relative to the CBSA in 2011."
    ) +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank(),
        text = element_text(size = 11, family = "Times")
    ) +
    scale_fill_brewer(palette = "Spectral") +
    guides(fill = guide_legend(title = "Retail Category"))

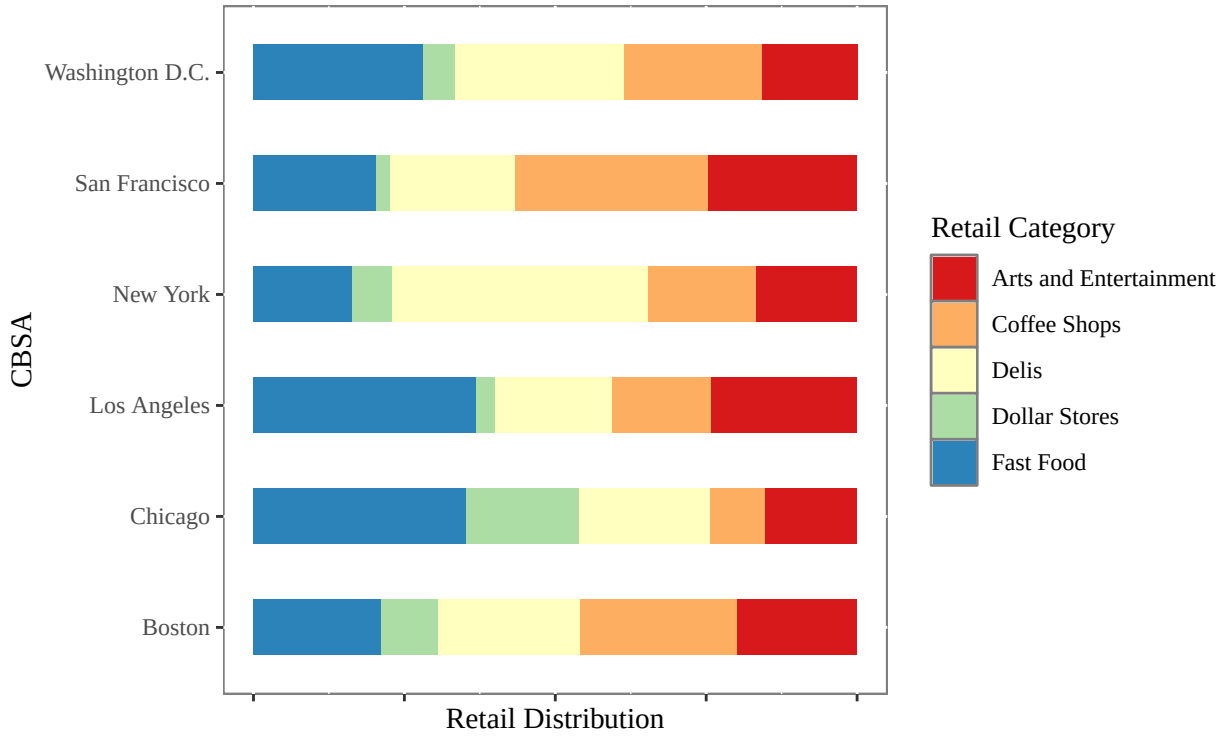
mod_chart <- ggplot(
    grouped_med,
    aes(y = cbsa_name, x = count, fill = category_pretty)
) +
    geom_bar(stat = "identity", position = "fill", width = 0.5) +
    xlab("Retail Distribution") +
    ylab("CBSA") +
    labs(
        title = "Figure 3: Moderate-Income Retail Distribution by CBSA in 2011",
        caption = "Zip codes with a median income below the top twenty-fifth \n
        ↪  percentile and above the fortieth percentile relative to the CBSA in 2011."
    ) +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank(),
        text = element_text(size = 11, family = "Times")
    ) +
    scale_fill_brewer(palette = "Spectral") +
    guides(fill = guide_legend(title = "Retail Category"))

# TODO: plot individual MSAs by low, med, hi

ggsave(
    "4_drafts/figures/retail_hi.png",
```

```
    hi_chart,
    width = 170,
    height = 100,
    units = "mm"
)
ggsave(
    "4_drafts/figures/retail_lo.png",
    lo_chart,
    width = 170,
    height = 100,
    units = "mm"
)
ggsave(
    "4_drafts/figures/retail_mod.png",
    mod_chart,
    width = 170,
    height = 100,
    units = "mm"
)
show(lo_chart)
```

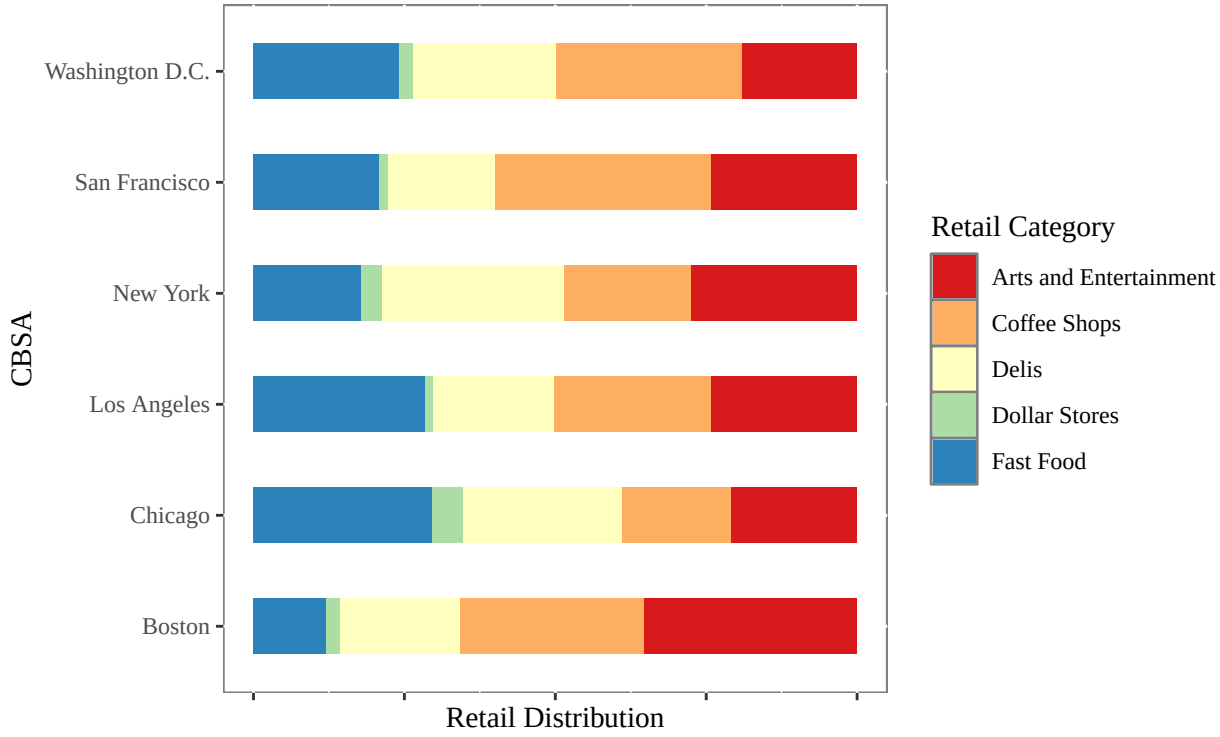
Figure 2: Low−Income Retail Distribution by CBSA in 2011



Zip codes with a median income in the bottom fortieth percentile relative to the CBSA in 2011.

```
show(mod_chart)
```

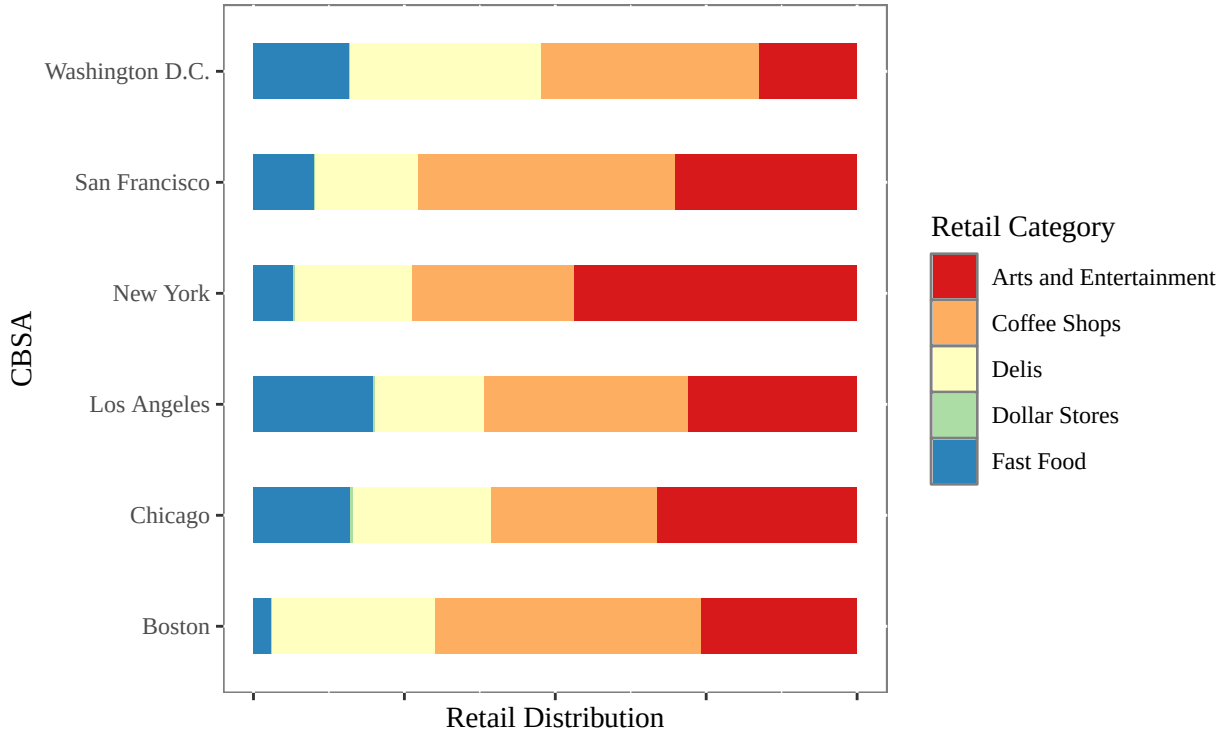
Figure 3: Moderate−Income Retail Distribution by CBSA in 2011



Zip codes with a median income below the top twenty−fifth percentile and above the fortieth percentile relative to the CBSA in 2011.

```
show(hi_chart)
```

Figure 4: High−Income Retail Distribution by CBSA in 2011



Zip codes with a median income in the top twenty−fifth percentile relative to the CBSA in 2011.